

Write a python program to implement of RSA Algorithm



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Asymmetric-Key Cryptography

- **Asymmetric-key** cryptography uses two keys: one private and one public
 - Also called public key cryptography
 - Named for its inventors Rivest, Shamir, and Adleman (RSA).
- For n users:
 - $\frac{n(n-1)}{2}$ secrets are required for symmetric-key cryptography
 - Only n personal secrets are needed in asymmetric-key cryptography
- It requires keys of at least 2048 bits for good security (versus 256 bits for symmetric- key algorithms)
 - Hence it is quite slow
- Asymmetric-key cryptography is normally used to encrypt or decrypt small pieces of information



Symmetric-key vs Asymmetric-key ciphers

Symmetric-key Cipher	Asymmetric-key Cipher
Symmetric-key cryptography is based on sharing secrecy	Asymmetric-key cryptography is based on personal secrecy
For n people, $n(n-1)/2$ shared secrets are needed	For n people, only n personal secrets are needed
Symmetric-key cryptography is based on substitution and permutation of symbols (characters or bits)	Asymmetric-key cryptography is based on applying mathematical functions to numbers
The plaintext and ciphertext are thought of as a combination of symbols.	The plaintext and ciphertext are numbers
Encryption and decryption permute these symbols or substitute one symbol for another	Encryption and decryption are mathematical functions that are applied to numbers to create other numbers



Asymmetric-key cryptosystem

- Asymmetric key cryptography uses two separate keys: one private and one public.
- Encryption** $\Rightarrow$ using **Public Key**
- Decryption** $\Rightarrow$ using **Private key**

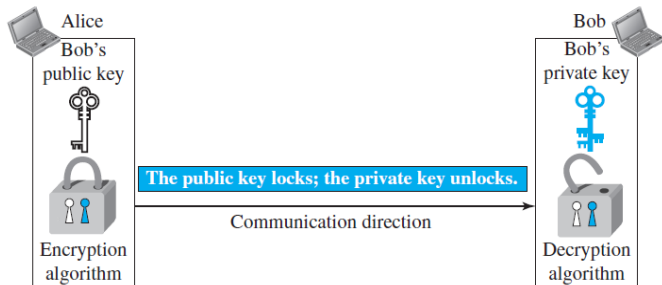


Figure: Credits: Forouzan. Locking and unlocking in asymmetric-key cryptosystem



Encryption and Decryption

- Choose two very large prime numbers p and q
 - Prime number is one that can be divided only by 1 and itself
- Find $n = p \times q$
- Find $\phi(n) = (p - 1) \times (q - 1)$
- Choose e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$
- Find d such that $(e \times d) \bmod \phi(n) = 1$.
- Public key (e, n) ; Private key (d, n)
- Encrypt: $C = (P^e) \bmod n$
- Decrypt: $P = (C^d) \bmod n$
- **P is always in the form of numbers**
 - If text is given, use ASCII values of each character



Example 1

The RSA algorithm uses two prime numbers, say $p=3$ and $q=11$. Discover the possible values of Public Key(e, n), and Private Key (d, n). Encrypt the message $P=4$;



Example 1: Finding Keys

- Let us choose $p = 3$; $q = 11$;
- $n = 3 \times 11 = 33$
- $\phi(n) = (p - 1) \times (q - 1) = 2 \times 10 = 20$
- Choose $e = 3$; $\gcd(3, 20) = 1$;
- Find d such that $(3 \times d) \bmod 20 = 1 \implies d = 7$
- Public Key: $(e = 3, n = 33)$
- Private Key: $(d = 7, n = 33)$



Example 1: Encryption and Decryption

- Public Key: $(e = 3, n = 33)$
- Private Key: $(d = 7, n = 33)$
- Message $P=4$
- Encrypt: $C=(P^e) \bmod n=(4^3) \bmod 33=64 \bmod 33=31$
- Decrypt: $P=(C^d) \bmod n=(31^7) \bmod 33=4$



Example 2: Finding Keys

- Let us choose $p = 7$; $q = 11$;
- $n = 7 \times 11 = 77$
- $\phi(n) = (p - 1) \times (q - 1) = 6 \times 10 = 60$
- Choose $e = 13$; $\gcd(13, 60) = 1$;
- Find d s.t $13 \times d \bmod 60 = 1 \implies d = 37$
- Public Key: $(e = 13, n = 77)$
- Private Key: $(d = 37, n = 77)$



Example 2: Encryption and Decryption

- Public Key: $(e = 13, n = 77)$
- Private Key: $(d = 37, n = 77)$
- Message $P=5$
- Encrypt: $C=(P^e) \bmod n=(5^{13}) \bmod 77=26 \bmod 77=26$
- Decrypt: $P=(C^d) \bmod n=(26^{37}) \bmod 77=5$



Finding Keys: Example 3

- Let us choose $p = 17$; $q = 11$;
- $n = 17 \times 11 = 187$
- $\phi(n) = (p - 1) \times (q - 1) = 16 \times 10 = 160$
- Choose $e = 7$
- Find d s.t $7 \times d \bmod 160 = 1 \implies d = 23$
- Public Key: $(e = 7, n = 187)$
- Private Key: $(d = 23, n = 187)$
- Find the plain text from the Ciphertext=65